STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 8B

Description

8.b) Write R program to find Analysis and Covariance with the sample data and visualize the regression graphically.

Procedure and Overvation:

Aim:

Write R program to find analysis and covariance with the sample data and visualize the regression graphically.

Description:

We use regression analysis to create models which describe the effect of variation in predictor variables on the res
ponse variables.

Sometimes if we have a categorical variable with values like yes or no or male or female etc...

The simple regression analysis gives multiple results for each value of the categorical variable by using it along w
ith the predictor variable

and comparing the regression lines for each level of the categorical variable.Such an analysis is termed as analysis
covariance also called as 'ANCOVA'.

Regression:

Regression is a statistical tool used to find a relationship between 2 variables.

In linear regression these variables are related through a general equation.

y=ax+b

where

y-> response variable

x-> predictor variable

a,b->constants

To find the linear regression we use a function lm()

SYNTAX:

lm(formula,Data)

formula is the relationship between x and y

Data is the set on which formula will be applied.

Predict:

SYNTAX

predict(object,newdata)

ANCOVA- Analysis of covariance

We create regression model taking hp as the predictor variable and mpg as the response variable taking into account
 the interaction between am and hp

SOURCE CODE:

```
input<-mtcars[,c("am","mpg","hp")]

print(head(input))

input<-mtcars

result<-aov(mpg~hp*am,dat=input)

print(summary(result))
```

```
#create the regression models

result1<-aov(mpg~hp*am,data=input)

result2<-aov(mpg~hp+am,data=input)

#compare the two models

print(anova(result1,result2))
```

```
11:1   (Top Level) ÷                                   R Script ÷
R 4.2.0 · ~/
>
> source("C:/Users/student/Desktop/20R01A6753/week8b.r")
                   am  mpg  hp
Mazda RX4           1 21.0 110
Mazda RX4 Wag       1 21.0 110
Datsun 710          1 22.8  93
Hornet 4 Drive      0 21.4 110
Hornet Sportabout   0 18.7 175
Valiant             0 18.1 105
            Df Sum Sq Mean Sq F value   Pr(>F)
hp           1  678.4   678.4  77.391 1.50e-09 ***
am           1  202.2   202.2  23.072 4.75e-05 ***
hp:am        1    0.0     0.0   0.001    0.981
Residuals   28  245.4     8.8
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Analysis of Variance Table

Model 1: mpg ~ hp * am
Model 2: mpg ~ hp + am
  Res.Df    RSS Df  Sum of Sq     F Pr(>F)
1     28 245.43
2     29 245.44 -1 -0.0052515 6e-04 0.9806
> |
```

VIVA RESPONSES

Q) Data Analytics uses ___ to get insights from data.

Statistical methods

Q) Amongst which of the following is / are the branch of statistics which deals with the development of statistical methods is classified as ___.

Applied statistics

Q) Linear Regression is the supervised machine learning model in which the model finds the best fit ___ between the independent and dependent variable.

Linear line

Q) Amongst which of the following is / are the types of Linear Regression,

Both A and B

Q) Amongst which of the following is / are the true about regression analysis?

Modeling relationships within the data